# Follow-the-Expert: Expert-Preserving Demonstration Synthesis in a Gaussian-Splat Twin

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**Abstract:** A 3D Gaussian Splatting (3DGS) twin can turn one real demonstration into hundreds of rendered ones and train a policy that runs on the real robot. We find that for skills defined by their path, this Real2Sim2Real loop works best when the synthesized motion stays the expert's. Follow-the-Expert (FTE) replays each phase of one VR demonstration as a dynamic movement primitive, moves only the phase boundaries and their object splats, and lets the splat's density steer the gripper around obstacles. On a real Boston Dynamics Spot, FTE policies succeed in 75.6% of rollouts on four shape-sensitive tasks, against 40.0% for trajectory optimization and 3.8% for keyframe motion planning; all three tie at 91.3% where only the end state matters. FTE also leads with obstacles absent from the demonstration (86.7% vs. 60.0% on three tasks), and doubling the offset bounds needs only more rendered demonstrations.

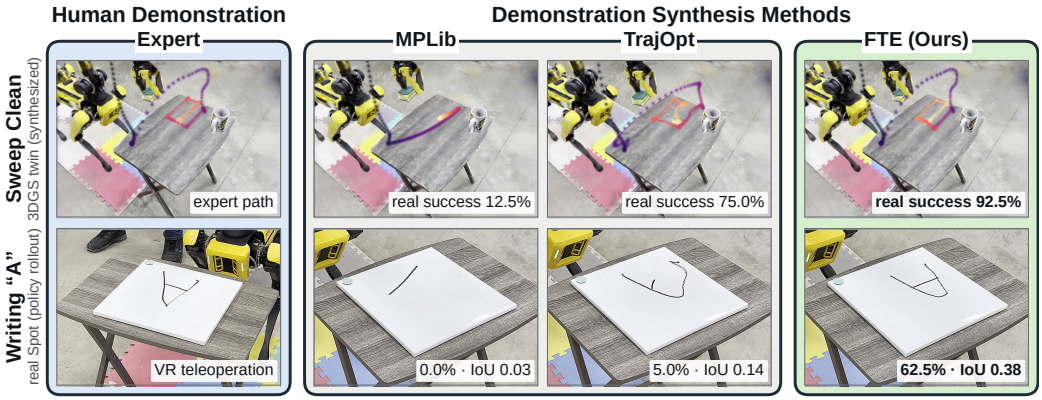



Figure 1: **Keeping the expert's motion is what transfers to the real robot.** Top: Sweep Clean demos synthesized in the 3DGS twin (end-effector path, dark to bright; Expert: the demo replayed in the twin) and each generator's real-robot success. Bottom: Writing "A" on the real Spot after one rollout per policy (Expert: the VR demonstration); success and mean stroke IoU over 40 rollouts.

## 1 Keep the expert, move the boundaries

**A splat twin makes demonstrations cheap; keeping the expert makes them right.** 3DGS reconstructions [1] train real-robot policies on rendered data [2, 3, 4]. One-demonstration generators in this family, RoboSplat [5] and 1001 DEMOS [6], retarget keyframes and plan or optimize the motion between them; this suits pick-and-place but replaces or distorts the expert's path when the path is the task (Fig. 1). We re-implement their generators as our baselines MPLib (keyframes plus motion planning) and TrajOpt (keyframes plus trajectory optimization). Segment-based generators keep object-centric segments found from contact [7, 8, 9]; FTE keeps every phase, including contact-free ones such as a camera scan, and uses the splat as a repulsive potential rather than a collision map [10].

Late-Breaking Results (non-archival), 10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

| | No obstacles | | | With obstacles absent from the demonstration | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Success ↑, 256 demos | | | Success ↑, 256 | | Success ↑, 500 | | Collision ↓, 500 | |
| **Task** | MPLib | TrajOpt | FTE | TrajOpt | FTE | TrajOpt | FTE | TrajOpt | FTE |
| Sweep Clean | 12.5 | 75.0 | **92.5** | 67.5 | **80.0** | 82.5 | **95.0** | 22.5 | **7.5** |
| Push–Pick–Place | 2.5 | 30.0 | **75.0** | 30.0 | **50.0** | 35.0 | **82.5** | 60.0 | **12.5** |
| Writing "A" | 0.0 | 5.0 | **62.5** | – | – | – | – | – | – |
| Videography | 0.0 | 50.0 | **72.5** | 42.5 | **60.0** | 62.5 | **82.5** | 27.5 | **22.5** |
| Sponge Pick–Place | **95.0** | 92.5 | 92.5 | 82.5 | **85.0** | 92.5 | **95.0** | **2.5** | **2.5** |
| Stove | 87.5 | **90.0** | **90.0** | 77.5 | **80.0** | 95.0 | **97.5** | 5.0 | **2.5** |

Table 1: **Real-robot success and collision rates (%), 40 rollouts per cell**; one policy per cell, trained on 1 real + 256 or 500 synthetic demos. Shape-sensitive tasks above the rule; bold: best per group; –: not run. Success by inspection (Writing: two judges, 85–100% agreement); collision: visible contact. MPLib plans without obstacle geometry, as in RoboSplat (10.0–17.5% at 500 with a 3DGS proxy).

**FTE moves the boundaries and replays the motion.** One handheld scan of the lab (reconstructed in 5–10 min, shared by all tasks) is registered to the robot by ICP (7.7 mm trimmed-mean residual); Spot's own splat, split into per-link Gaussians, is posed by forward kinematics. One VR demonstration is split at operator button presses into phases, each fitted with Cartesian and orientation dynamic movement primitives (DMPs) [11]; shifting every press by $\pm 0.2$ s changes the synthesized paths' mean DTW to the expert by at most 1.5 mm on three tasks. During synthesis we sample bounded offsets of the phase boundaries ($1\times$ bounds, about 0.1 m per axis), move the linked object splats with them, and roll out each DMP with the expert's forcing term [12], keeping the approach, contact path, gripper timing, and orientation. Near obstacles, the summed density $\rho$ of designated obstacle Gaussians (an uncalibrated proxy) pushes the end effector along $-\nabla\rho$ before the rollout rejoins the expert's path [13]. Rendered front and top views plus the real demo train a Diffusion Policy [14]; the twin simulates no dynamics. Each task then needs only a bounds edit (2–3 min) and one demonstration (1–2 min), plus a scan (about 10 min) for a new object; no code changes.

## 2 Three findings on a real Spot

**1. Keeping the path decides the shape-sensitive tasks.** Four tasks are shape-sensitive: wiping a spill (Sweep Clean), pushing a block before picking it (Push–Pick–Place), writing "A", and filming a flower along an arc (Videography, six phases); Sponge Pick–Place and Stove are endpoint controls. Only the generator differs: all methods share the twin, demonstration, policy, and 40 test configurations inside the $1\times$ bounds. FTE leads TrajOpt on every shape-sensitive task (Table 1), 121/160 rollouts against 64/160 (MPLib 6/160; task-stratified Cochran–Mantel–Haenszel (CMH) test, $p < 10^{-10}$), and all three reach 91.3% on the endpoint controls. Path-preserving warps match FTE on Sweep Clean (affine 87.5%, spline 92.5%): the path is what matters, and DMPs give obstacle avoidance an acceleration-level entry point.

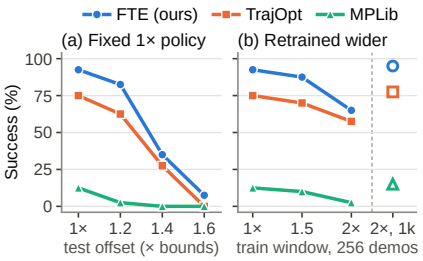



Figure 2: **Range costs rendering.** Sweep Clean, $N$=40 per point. (a) $1\times$ policies tested farther out. (b) Retrained on wider windows; hollow: $2\times$ with 1,000 demos.

**2. The splat that renders the views also clears obstacles.** With an obstacle absent from the demo, FTE leads TrajOpt on the three shape-sensitive tasks at both budgets (86.7% vs. 60.0% at 500 demos, CMH $p < 10^{-5}$) and collides less on each; without the density term, FTE drops to 0–17.5% on Sweep, Push–Pick–Place, Sponge, and Stove.

**3. Range costs rendering, not teleoperation.** Beyond its bounds a fixed $1\times$ policy degrades with the baselines (Fig. 2a), since FTE retargets locally. Retrained on a $2\times$ window, 256 demos spread too thin (Sweep 65.0%, Fig. 2b; Sponge 67.5%); 1,000 reach 95.0% and 97.5%, back at the $1\times$ level.

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